ECEN 104 – Electromagnetics I

Lab 6: Magnetic Resonance Imaging (MRI)

Group I

1. Introduction

Magnetic resonance imaging (MRI) is one of the most advanced noninvasive medical imaging techniques, relying on the excitation of proton magnetic moments through biasing magnetic fields (B_0), gradient fields, and radio-frequency (RF) pulse fields. This lab explores three primary components of MRI field generation: the uniform biasing field, the linear gradient field, and the birdcage RF excitation field. By constructing computational models for each configuration, the experiment demonstrates how magnetic field uniformity and directionality are achieved using static current distributions.

The primary goals of this lab were to:
1. Visualize and quantify the uniform biasing field (B_0) using a Maxwell coil configuration.
2. Generate and analyze the z-axis gradient field using a pair of oppositely driven coils.
3. Model the transverse magnetic field of a birdcage coil and study how the number of elements (N) and phase affect field uniformity and orientation.

The overall findings show that the Maxwell coil produces a highly uniform B_0 field near its center, the gradient coil introduces a controlled linear variation along the z-axis, and the birdcage coil creates an approximately uniform transverse field whose direction depends on current phasing.

2. Hypothesis

It is hypothesized that increasing the number of coils or current elements within each MRI coil system will improve magnetic field uniformity within the imaging region. Specifically, a Maxwell coil arrangement will produce a nearly homogeneous B_0 field along its axis; the gradient coil will exhibit a linear magnetic field variation between coils; and a birdcage coil with higher N will yield a more uniform transverse field whose direction can be rotated by phase-shifting the current distribution.

3. Procedure and Analysis

3.1 Part 1 – The B_0 Field (Maxwell Coil)

A Maxwell coil configuration consists of three coaxial circular coils: a central coil at the origin and two outer coils placed symmetrically at ±d along the z-axis, each carrying current I in the same direction. This setup minimizes the third derivative of the magnetic field, resulting in high homogeneity at the center.

The on-axis magnetic field of a single loop is given by:

$$B_z(z) = (\mu_0 * I * a^2) / (2 * (a^2 + z^2)^{(3/2)}) \quad \text{(Eq. 1)}$$

where μ_0 is the permeability of free space, I is the coil current, a is the radius, and z is the axial distance. For the Maxwell coil, the total on-axis field is the superposition of the three coils:

$$B_{total}(z) = B_z(z - d) + B_z(z) + B_z(z + d) \quad \text{(Eq. 2)}$$

The center coil radius was specified to be R = 1.5 m; it is known for a Maxwell coil that the outer coil radius is given by sqrt(4/7)R and distance by sqrt(3/7)R. The currents were scaled to produce a maximum field of 1 T at the center in figure 1. Field visualizations were generated using the provided `loopBfield` MATLAB function for figure 2..

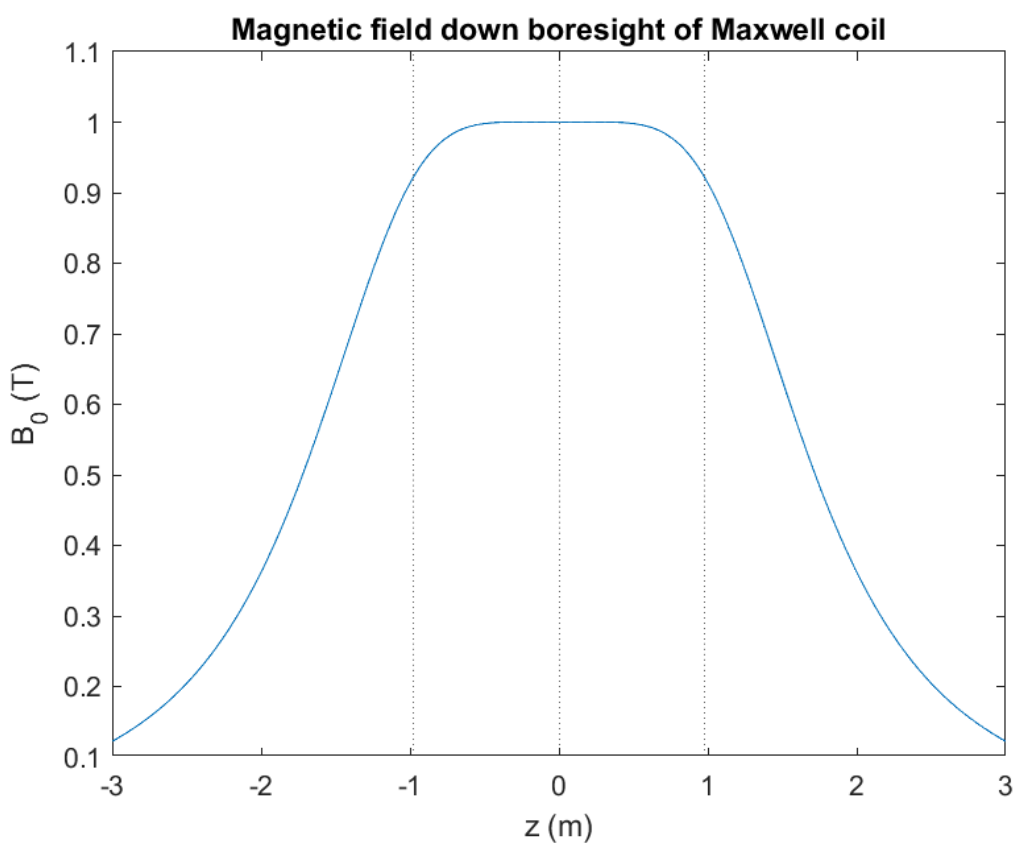

Figure 1: Simulated magnetic field magnitude for Maxwell coil

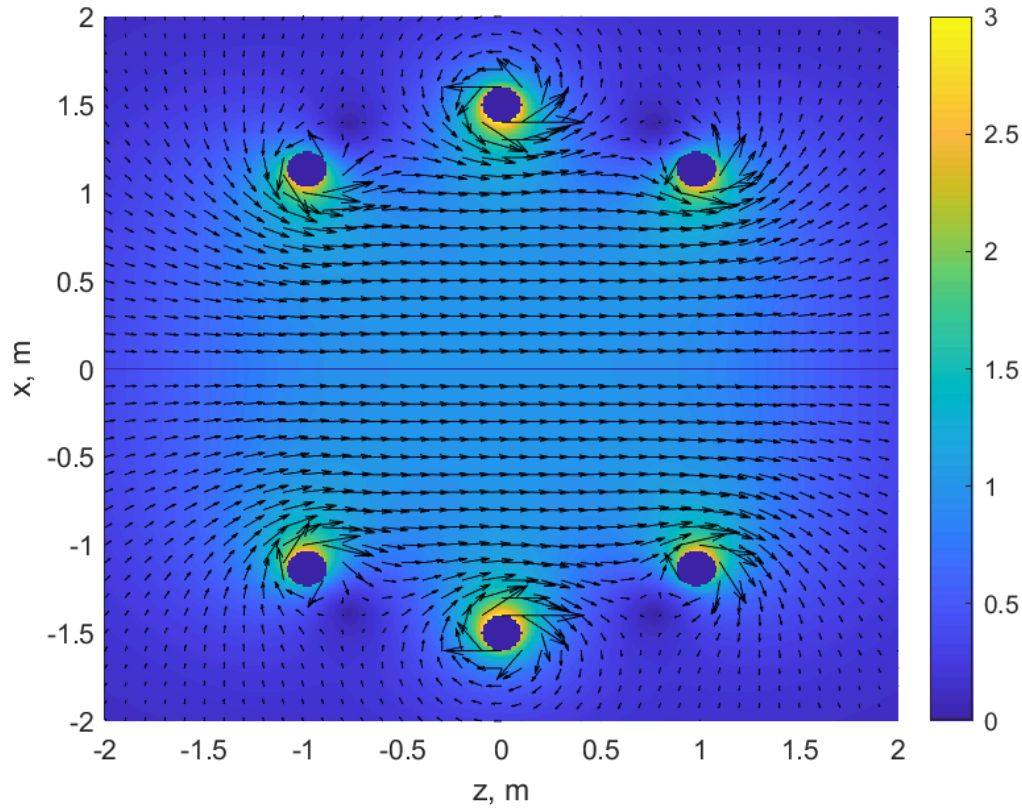

Figure 2: Quiver Plot of Magnetic Field

From the results, the magnetic field was observed to be relatively homogeneous near the center region. However, field gradients increased farther from the center. Increasing the number of turns, adjusting coil spacing, or using higher-order correction coils (e.g., Helmholtz or actively shielded coils) can further improve uniformity.

3.2 Part 2 – Z-Gradient Coil

Gradient coils introduce controlled variations in the magnetic field, enabling spatial encoding of signals in MRI. In this experiment, two circular coils of radius a = 0.75 m were separated by 1.5 m and driven with equal and opposite currents to produce a linear variation of B_z along the z-axis.

The magnetic field along the axis for a single loop follows Eq. (1). The superposition of the two oppositely driven loops yields:

$$B_{gradient}(z) = B_z(z - d/2) - B_z(z + d/2) \quad \text{(Eq. 3)}$$

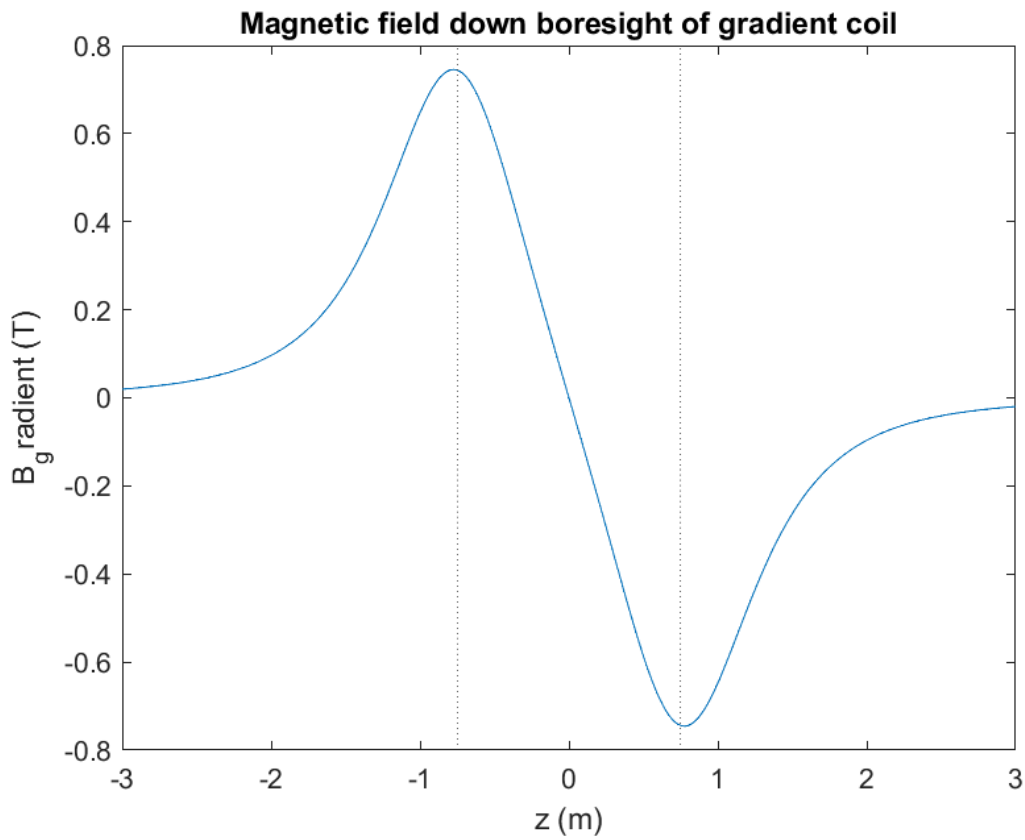

Figure 3: On-axis magnetic field for z-gradient coil

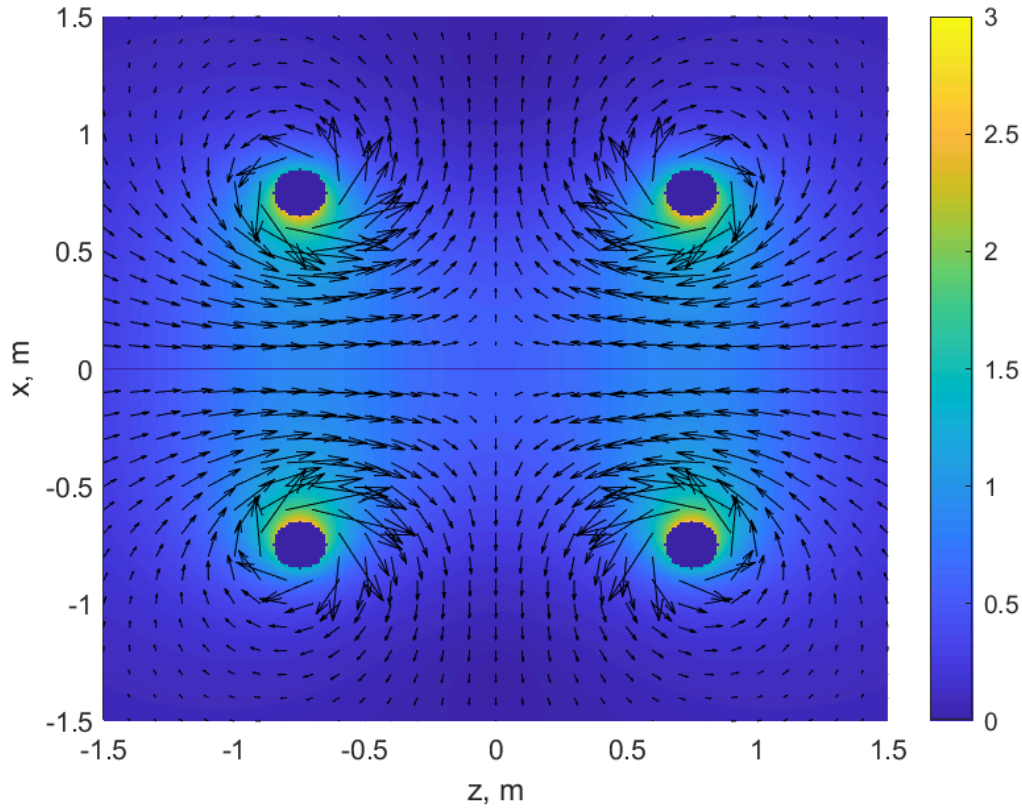

Figure 4: Quiver Plot of Magnetic Field

Looking at figure 3, the results showed that the field magnitude changes linearly between the coils (approximately from z = -0.75 m to z = 0.75 m), while nonlinearities appear near and beyond the coil edges. This indicates that the gradient is most uniform near the center of the coil pair. As observed in figure 4, the direction of the gradient field aligns with the z-axis, with the slope of $B_z(z)$ dependent on coil separation and current magnitude.

3.3 Part 3 – Birdcage Coil

Birdcage coils are used to generate RF magnetic fields (B_1) orthogonal to the main B_0 field. The coil consists of N evenly spaced vertical current elements arranged around a circular frame of radius a. The current magnitude on each element varies sinusoidally as:

$$I_n = I_0 * \sin(\varphi_n) \quad \text{(Eq. 4)}$$

where $\varphi_n = 2\pi n/N$ is the angular position of the nth element. The resulting interior fields in figures 5, 6, and 7 are nearly uniform and directed along the x-axis. As N increases, field uniformity improves due to smoother spatial current sampling around the circumference. Field visualizations were generated using the provided `lineBfield` MATLAB function.

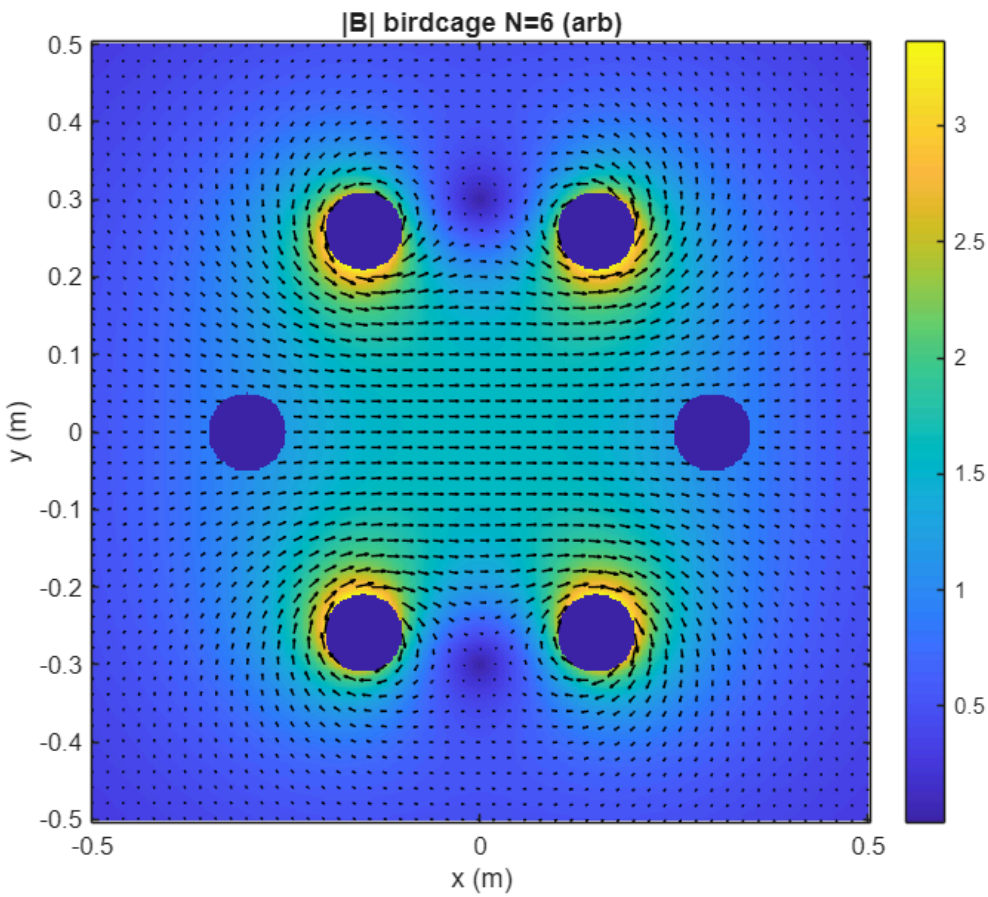

Figure 5: Quiver Plot of magnetic field for 6 birdcage coils

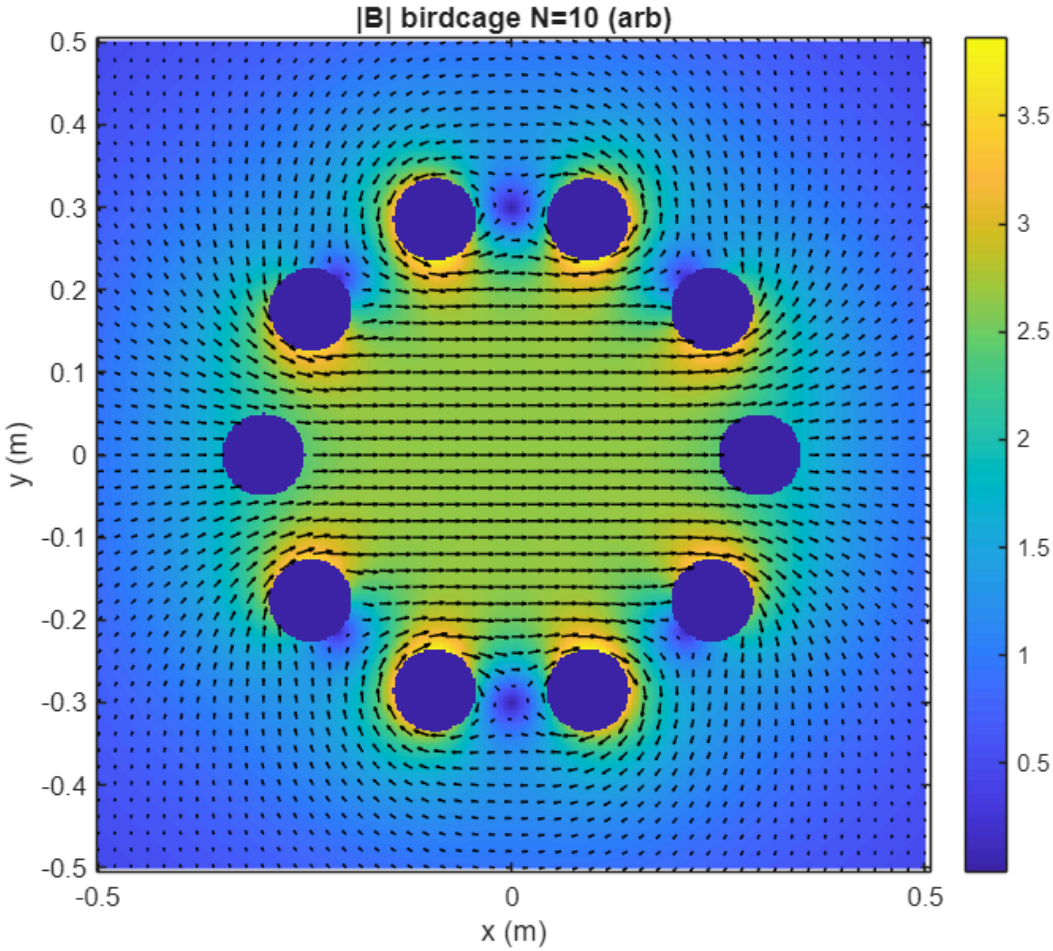

Figure 6: Quiver Plot of magnetic field for 10 birdcage coils

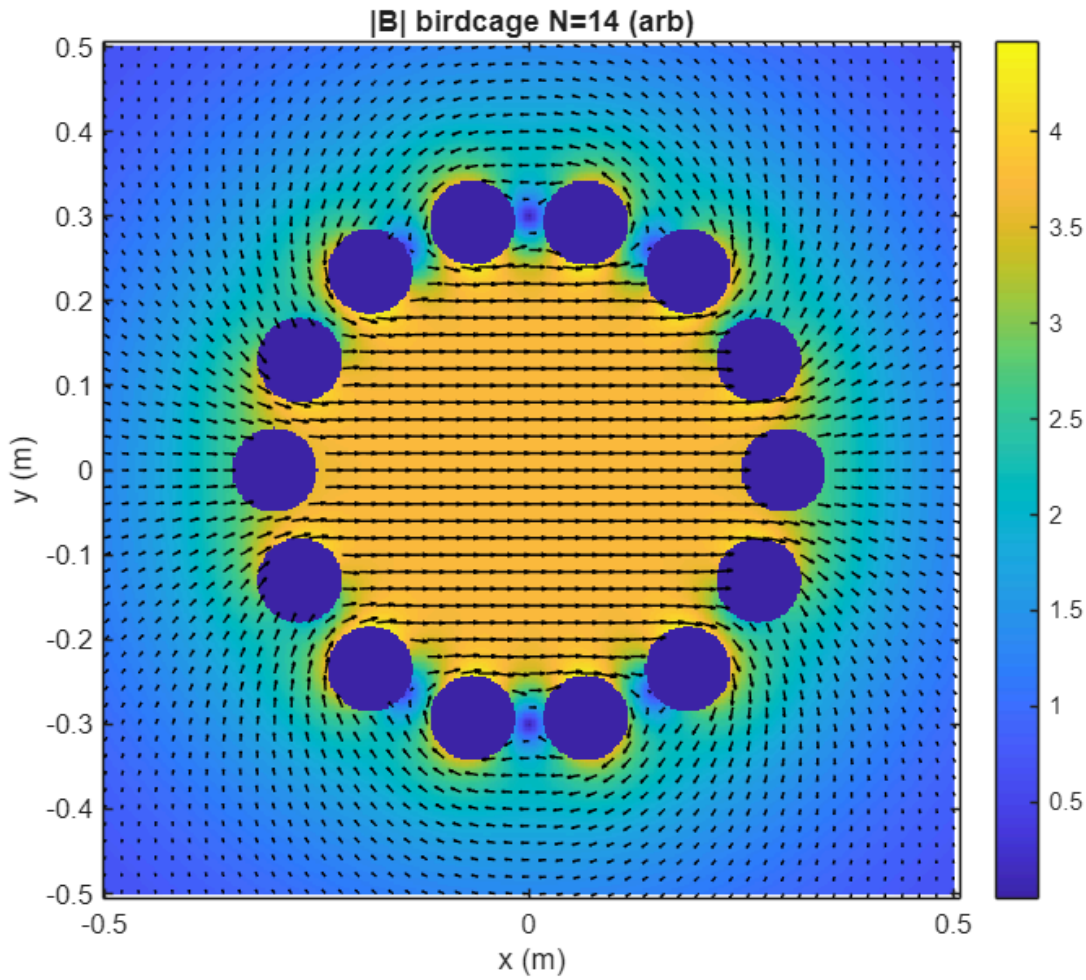

Figure 7: Quiver Plot of magnetic field for 14 birdcage coils

Experimental modeling confirmed that the field points in the x-direction for the given sine distribution. Increasing N from 6 to 14 elements improved homogeneity considerably. Additionally, as shown in figure 8, phase-shifting the current distribution by π/2 rotates the interior magnetic field by 90°, enabling control over field orientation for different imaging modes.

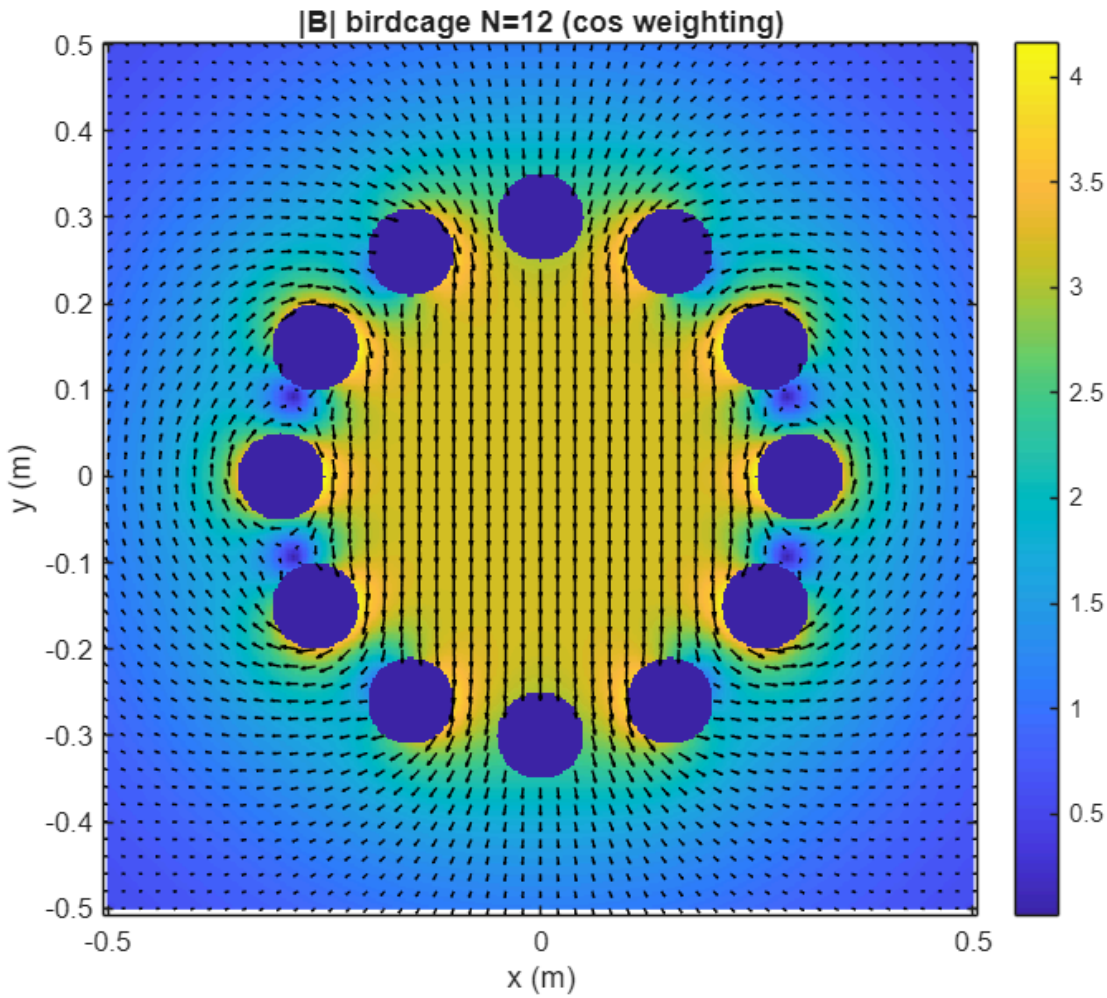

Figure 8: Quiver Plot of magnetic field for 12 birdcage coils with cos weighting

4. Conclusions

This lab demonstrated the magnetic field configurations central to MRI operation: biasing (B_0), gradient, and RF (B_1) fields. The Maxwell coil produced a nearly uniform field suitable for spin alignment, the z-gradient coil achieved a controllable linear variation enabling spatial encoding, and the birdcage coil generated a uniform transverse RF field essential for excitation.

Key findings include:

• The Maxwell coil achieved a high degree of uniformity near its center, with slight edge

distortion.
• The z-gradient coil showed a nearly linear change in magnetic field between coils, suitable for encoding spatial position.
• The birdcage coil produced increasingly homogeneous fields with higher element counts (N), and its orientation could be rotated by shifting the phase of the applied currents.

Future improvements could involve using multipole expansions or numerical optimization to refine coil design for improved field homogeneity and reduced fringe distortion. These analyses form the foundation for understanding the physical principles underlying MRI hardware design and spatial encoding.

5. References

[1] Wikipedia. "Physics of Magnetic Resonance Imaging." Available: https://en.wikipedia.org/wiki/Physics_of_magnetic_resonance_imaging
[2] MRI Questions. "Gradient Coils." Available: http://mriquestions.com/gradient-coils.html
[3] MRI Questions. "Birdcage Coil." Available: http://mriquestions.com/birdcage-coil.html